THE BOOK RAMSEY NUMBER $R(B_{66}, B_{67})$

DEEP BHATTACHARJEE

Abstract. We prove that $R(B_{66}, B_{67}) = 267$, the case $n = 67$ of Wesley's conjecture $R(B_{n-1}, B_n) = 4n - 1$, which no earlier construction covered. The extremal graph is a two-block circulant graph over $\mathbb{Z}_{133}$ whose connection sets are unions of orbits of the cube roots of unity modulo 133.

1. Introduction

The book B_k is the graph $K_2 + \overline{K_k}$: k triangles sharing one edge, its spine. For graphs G and H, the Ramsey number $R(G, H)$ is the least N such that every graph on N vertices contains G or has a complement containing H. Rousseau and Sheehan [7] proved, from Goodman's count of monochromatic triangles [4], that

$$R(B_{n-1}, B_n) \le 4n - 1 \qquad (n \ge 2). \tag{1.1}$$

Wesley [10] proved equality for $n \le 20$ and whenever $2n - 1$ is a prime power congruent to 1 modulo 4, and conjectured equality for every n. The conjecture is one of the open problems of Epoch AI's FrontierMath collection [3], which records further progress. Turturean [9] constructed extremal graphs for all $n \le 56$ and for fifteen further values below 100, partly through a cyclotomic construction of Szekeres [8] that applies when $4n - 1$ is a prime congruent to 3 modulo 8; Chung, Du and Wesley [2] found three further infinite families, from conference graphs, from Legendre pairs and from Yamada–Pott designs; and Price obtained $n = 70$ [3].[1] The value $n = 67$ was not among them. We settle it.

Theorem. *$R(B_{66}, B_{67}) = 267$.*

The upper bound is (1.1). For the lower bound we exhibit a graph on 266 vertices with no B_{66} whose complement has no B_{67}. It is a two-block circulant graph over $\mathbb{Z}_{133}$, the shape used by Wesley [10] and Turturean [9], and its connection sets are invariant under multiplication by the nine cube roots of unity modulo 133. Section 2 reduces the two book conditions to one inequality (2.1) on the connection sets; the symmetry then reduces (2.1) to ten cases, which Table 1 records. Every step can be checked by hand, and the graph has also been checked by two independent programs and in Lean 4 (see the end of the paper).

2. Two-block circulant graphs

Throughout, $n \ge 2$, $v = 2n - 1$ and $G = \mathbb{Z}_v$. For $X, Y \subseteq G$ and $d \in G$ put

$$\mathrm{N}_{X,Y}(d) = |X \cap (Y + d)| = \#\{y \in Y : \ y + d \in X\}.$$

Translating by $-d$ shows $\mathrm{N}_{X,X}(-d) = \mathrm{N}_{X,X}(d)$, and $\sum_{d \in G} \mathrm{N}_{X,Y}(d) = |X|\,|Y|$.

2020 *Mathematics Subject Classification.* Primary 05C55; Secondary 05C25, 05B10.

Key words and phrases. Ramsey number, book graph, block-circulant graph, multiplier group.

Corresponding author: d.bhattacharjee@erl-forschung.de; ORCID 0000-0003-0466-750X.

[1] The values $n \le 100$ that none of the constructions cited here covers are 58, 59, 60, 64, 65, 67, 68, 72, 73, 76, 78, 80, 81, 84, 88, 89, 92, 93, 94, 96 and 100. This list is our reading of [2, 3, 9, 10] as of 14 October 2026.

Let $A \subseteq G \setminus \{0\}$ with $A = -A$ and $|A| = n-1$, let $C = G \setminus (A \cup \{0\})$, so that $C = -C$ and $|C| = n-1$, and let $B \subseteq G$ with $|B| = n-1$. The graph $\Gamma = \Gamma(A, B)$ has vertex set $G \times \{0,1\}$, and

$$(x,0) \sim (y,0) \iff y - x \in A, \qquad (x,1) \sim (y,1) \iff y - x \in C,$$

and $(x,0) \sim (y,1)$ if and only if $y - x \in B$. Since A and C are symmetric and avoid 0, Γ is a simple graph; in the notation of [10] it is $\Gamma_G(D_{11}, D_{12}, D_{22})$ with $D_{11} = A$, $D_{12} = B$, $D_{22} = C$. A vertex $(x,0)$ has $|A| + |B| = 2n-2$ neighbours and a vertex $(x,1)$ has $|C| + |B| = 2n-2$, so Γ is $(2n-2)$-regular on $4n-2$ vertices. The translations $(x,i) \mapsto (x+t, i)$ are automorphisms. Write $\lambda(p,q)$ for the number of common neighbours of two vertices, $[\,P\,]$ for the indicator of a statement P, and

$$f(d) = \mathrm{N}_{A,A}(d) + \mathrm{N}_{B,B}(d) \qquad (d \in G).$$

Lemma 2.1. *For $d \neq 0$,*

$$\lambda\big((0,0),(d,0)\big) = f(d), \qquad \lambda\big((0,1),(d,1)\big) = f(d) - 1 + 2[\,d \in A\,],$$

and for every $d \in G$, $\lambda\big((0,0),(d,1)\big) = n - 1 - [\,d \in B\,]$.

Proof. A vertex $(z,0)$ is a common neighbour of $(0,0)$ and $(d,0)$ when $z \in A$ and $z - d \in A$, and $(z,1)$ is one when $z \in B$ and $z - d \in B$; this gives $\mathrm{N}_{A,A}(d) + \mathrm{N}_{B,B}(d)$.

For $(0,1)$ and $(d,1)$, the common neighbours $(z,1)$ are counted by $\mathrm{N}_{C,C}(d)$. A vertex $(z,0)$ is adjacent to $(0,1)$ when $-z \in B$ and to $(d,1)$ when $d - z \in B$; with $w = -z$ these say $w \in B$ and $w + d \in B$, so there are $\mathrm{N}_{B,B}(-d) = \mathrm{N}_{B,B}(d)$ of them. To compute $\mathrm{N}_{C,C}(d)$, let $S = A \cup \{0\}$, so $|S| = n$ and $C = G \setminus S$. Then $|C \cap (C+d)| = v - |S \cup (S+d)| = v - 2n + |S \cap (S+d)|$. The set $S \cap (S+d)$ is the disjoint union of $A \cap (A+d)$, of $\{0\}$ when $-d \in A$, and of $\{d\}$ when $d \in A$ (neither 0 nor d lies in $A \cap (A+d)$, because $0 \notin A$). As $A = -A$ and $v - 2n = -1$, we get $\mathrm{N}_{C,C}(d) = \mathrm{N}_{A,A}(d) - 1 + 2[\,d \in A\,]$.

For $(0,0)$ and $(d,1)$, a vertex $(z,0)$ is a common neighbour when $z \in A$ and $d - z \in B$, that is, $z = d - b$ with $b \in B$ and $d - b \in A$; since $A = -A$ the last condition reads $b - d \in A$. A vertex $(z,1)$ is a common neighbour when $z \in B$ and $z - d \in C$. Together, $\lambda = \#\{b \in B :\ b - d \in A \cup C\} = \#\{b \in B :\ b \neq d\} = n - 1 - [\,d \in B\,]$. □

Lemma 2.2. *In a $(2n-2)$-regular graph on $4n-2$ vertices, two distinct non-adjacent vertices p, q have exactly $\lambda(p,q)$ common non-neighbours other than p and q.*

Proof. The neighbourhoods of p and q lie in the other $4n-4$ vertices and their union has $2(2n-2) - \lambda(p,q)$ elements; the remaining $\lambda(p,q)$ vertices are adjacent to neither. □

Proposition 2.3. *$\Gamma(A,B)$ contains no B_{n-1} and its complement contains no B_n if and only if*

$$f(d) \leq n - 1 - [\,d \in A\,] \qquad \text{for every } d \in G \setminus \{0\}. \tag{2.1}$$

In that case $R(B_{n-1}, B_n) = 4n-1$.

Proof. An edge in k triangles is the spine of a B_k. So Γ contains no B_{n-1} if and only if every edge has at most $n-2$ common neighbours, and by Lemma 2.2 the complement contains no B_n if and only if every pair of distinct non-adjacent vertices has at most $n-1$ common neighbours. Both conditions together say $\lambda(p,q) \leq n - 1 - [\,p \sim q\,]$ for all $p \neq q$, and by translation it suffices to take $p = (0,i)$. We use Lemma 2.1. For $p = (0,0)$, $q = (d,1)$ we have $p \sim q$ if and only if $d \in B$, so the condition holds with equality; the pairs $(0,1), (d,0)$ are translates of these. For $p = (0,0)$, $q = (d,0)$ with $d \neq 0$ we have $p \sim q$ if and only if $d \in A$, and the condition is (2.1). For $p = (0,1)$, $q = (d,1)$ we have $p \sim q$ if and only if $d \notin A$, and the

condition $f(d) - 1 + 2[\,d \in A\,] \le n - 2 + [\,d \in A\,]$ is again (2.1). The last sentence follows from (1.1), since Γ has $4n-2$ vertices. □

Remark 2.4. Summing over $d \neq 0$ gives $\sum f(d) = 2(n-1)(n-2)$ and $\sum (n - 1 - [\,d \in A\,]) = (n-1)(2n-3)$. So when (2.1) holds, the slacks $s(d) = n - 1 - [\,d \in A\,] - f(d)$ are non-negative integers with sum $n-1$, and (2.1) is an equality for at least $n-1$ of the $2n-2$ values of d. The cross edges never matter: B enters only through its autocorrelation $\mathrm{N}_{B,B}$. For the Paley graphs of [10] (v a prime power, $A = B$ the nonzero squares), $s = 1$ on A and $s = 0$ on C.

Lemma 2.5. *If $u \in \mathbb{Z}_v^\times$ satisfies $uA = A$ and $uB = B$, then $f(ud) = f(d)$ and $[\,ud \in A\,] = [\,d \in A\,]$; moreover $f(-d) = f(d)$ and $[\,-d \in A\,] = [\,d \in A\,]$. Hence, if $H \le \mathbb{Z}_v^\times$ fixes A and B, it suffices to check (2.1) for one d in each orbit of the group $\pm H$ on $G \setminus \{0\}$.*

Proof. The map $y \mapsto uy$ is a bijection from $X \cap (Y + d)$ onto $uX \cap (uY + ud)$, so $\mathrm{N}_{uX,uY}(ud) = \mathrm{N}_{X,Y}(d)$. The statements about $-d$ follow from $\mathrm{N}_{X,X}(-d) = \mathrm{N}_{X,X}(d)$ and $A = -A$. □

3. The case $n = 67$

Let $n = 67$ and $v = 133 = 7 \cdot 19$. Let

$$H = \{u \in \mathbb{Z}_{133} : u^3 = 1\} = \{1, 11, 30, 39, 58, 64, 102, 106, 121\},$$

the group of cube roots of unity; under $\mathbb{Z}_{133} \cong \mathbb{Z}_7 \times \mathbb{Z}_{19}$ it is $\{1,2,4\} \times \{1,7,11\}$. Since $|H|$ is odd, $-1 \notin H$ and $\pm H$ has order 18. Its orbits on $\mathbb{Z}_{133} \setminus \{0\}$ are the following ten: the 108 units form six orbits of size 18, with representatives $1, 2, 3, 4, 6, 8$; on the 18 nonzero multiples of 7 the group acts through its image $\pm\{1,7,11\}$ modulo 19, which has order 6, giving three orbits, with representatives $7, 14, 28$; and on the six nonzero multiples of 19 it acts through its image $\pm\{1,2,4\} = \mathbb{Z}_7^\times$ modulo 7, giving one orbit, with representative 19. Put

$$A = \pm H \cdot \{1, 3, 4, 14, 28\},$$
$$B = H \cdot \{2, 5, 8, 10, 12, 14, 20, 28, 57, 70\}.$$

The first is a union of three orbits of size 18 and two of size 6, so $|A| = 66$ and $A = -A$; the second is a union of six H-orbits of size 9 and four of size 3, so $|B| = 66$. Explicitly,

$$\begin{aligned} A = \{&1, 3, 4, 9, 10, 11, 12, 13, 14, 16, 21, 23, 25, 27, 28, 30, 31, 33, 34, 35, 36, 39, 40,\\ &41, 42, 43, 44, 48, 52, 58, 59, 63, 64, 69, 70, 74, 75, 81, 85, 89, 90, 91, 92, 93, 94,\\ &97, 98, 99, 100, 102, 103, 105, 106, 108, 110, 112, 117, 119, 120, 121, 122, 123,\\ &124, 129, 130, 132\},\\ B = \{&2, 5, 8, 10, 12, 13, 14, 17, 18, 20, 21, 22, 24, 26, 27, 28, 31, 34, 37, 42, 45, 46, 48,\\ &50, 54, 55, 57, 60, 62, 63, 65, 68, 69, 70, 71, 73, 75, 78, 79, 83, 87, 88, 89, 91, 94,\\ &95, 96, 98, 103, 105, 107, 108, 109, 110, 111, 113, 114, 115, 116, 122, 124, 125,\\ &128, 129, 131, 132\}. \end{aligned}$$

The set B is fixed by H but not by -1: it meets $-B$ in 42 elements. Figure 1 draws both sets on the torus $\mathbb{Z}_7 \times \mathbb{Z}_{19}$.

Theorem 3.1. *The sets A and B above satisfy (2.1) for $n = 67$. Consequently $\Gamma(A, B)$ is a 132-regular graph on 266 vertices with no B_{66} whose complement has no B_{67}, and $R(B_{66}, B_{67}) = 267$.*

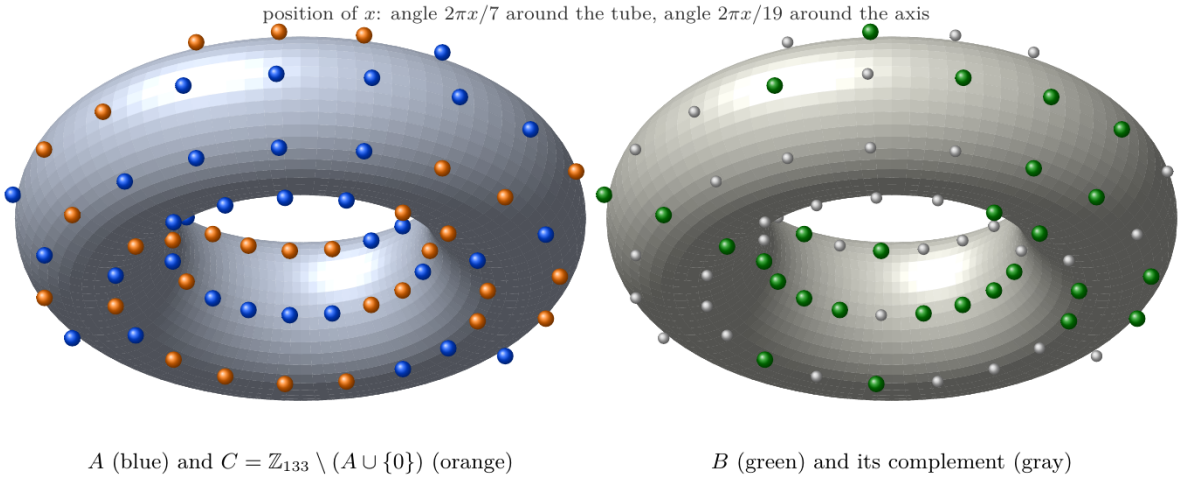


FIGURE 1. The sets A and B for $n = 67$. The residue $x \in \mathbb{Z}_{133}$ sits at angle $2\pi x/7$ around the tube and $2\pi x/19$ around the axis, so multiplication by $H = \{1, 2, 4\} \times \{1, 7, 11\}$ permutes the points of each picture within its colours.

TABLE 1. The inequality (2.1) for $n = 67$ on the ten orbits of $\pm H$. The last column is the slack $s(d)$ of Remark 2.4; the sizes times the slacks add up to 66.

d	$\lvert \pm Hd \rvert$	$d \in A$	$\mathrm{N}_{A,A}(d)$	$\mathrm{N}_{B,B}(d)$	$f(d)$	$66 - [\,d \in A\,]$	$s(d)$
1	18	yes	36	28	64	65	1
2	18	no	35	31	66	66	0
3	18	yes	30	35	65	65	0
4	18	yes	32	32	64	65	1
6	18	no	33	33	66	66	0
8	18	no	31	35	66	66	0
7	6	no	33	32	65	66	1
14	6	yes	30	33	63	65	2
28	6	yes	31	34	65	65	0
19	6	no	30	34	64	66	2

Proof. An element u of H fixes B, because B is a union of H-orbits, and fixes A, because A is a union of $\pm H$-orbits. By Lemma 2.5 it suffices to check (2.1) at the ten representatives, and Table 1 lists $\mathrm{N}_{A,A}(d)$ and $\mathrm{N}_{B,B}(d)$ there; each entry is the number of elements of a 66-element set whose translate by $-d$ stays in the set.[2] In every row $f(d) \le 66 - [\,d \in A\,]$. Proposition 2.3 gives the rest. □

[2]For instance $\mathrm{N}_{A,A}(1) = 36$ is the number of pairs $a - 1, a$ of consecutive residues in A: the runs $3, 4$; $9, \ldots, 14$; $27, 28$; $30, 31$; $33, \ldots, 36$; $39, \ldots, 44$; $58, 59$; $63, 64$; $69, 70$; $74, 75$; $89, \ldots, 94$; $97, \ldots, 100$; $102, 103$; $105, 106$; $119, \ldots, 124$; $129, 130$ contribute $1 + 5 + 1 + 1 + 3 + 5 + 1 + 1 + 1 + 1 + 5 + 3 + 1 + 1 + 5 + 1 = 36$.

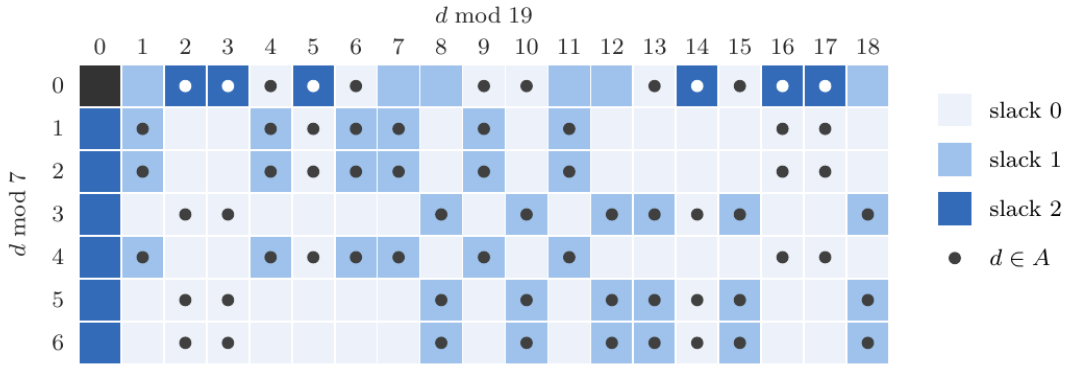


FIGURE 2. The slack $s(d) = 66 - [\,d \in A\,] - f(d)$ for $n = 67$ at every $d \neq 0$, with d placed by its residues modulo 7 and 19; the black cell is $d = 0$, and a dot marks $d \in A$. Equality holds for 78 values of d, the slack is 1 for 42 and 2 for 12.

4. REMARKS

Remark 4.1. The function f is not constant on A or on C (Figure 2): it takes the values 63, 64, 65 on A and 64, 65, 66 on C. So the example is not a difference-set construction of the kind behind the infinite families in [8, 9, 10], where f is constant on A and on C.

Remark 4.2. The example was found by a computer search. For fixed $v = 2n - 1$ and a subgroup $H \leq \mathbb{Z}_v^\times$, the sets A and B are taken to be unions of H-orbits, with $A = -A$; one 0/1 variable for each orbit in A and one for each orbit in B turn (2.1) into a system of quadratic inequalities, one for each H-orbit of d, together with $|A| = |B| = n - 1$. For $v = 133$ the cube roots of unity have 20 orbits on the nonzero residues, and the CP-SAT solver of OR-Tools [6] finds a solution in under a second. The same search did not produce examples for the other values in the footnote to the introduction, for the subgroups and the time limits we tried; some of these systems are infeasible. No step of the proof above depends on the search.

Question 4.3. Is there an infinite family of two-block circulant graphs, with connection sets invariant under a large multiplier group but f not constant on A, that settles further cases of $R(B_{n-1}, B_n) = 4n - 1$? The smallest open case is now $n = 58$, with $v = 115$.

Formal verification. Theorem 3.1 is also proved in Lean 4 [5], without any library beyond the Lean core. The development defines the graph $\Gamma(A, B)$ on the vertices $0, \ldots, 265$ by the two bit masks of A and B, and the kernel evaluates the common neighbours and common non-neighbours of all 35,245 pairs of vertices and checks that the adjacency is symmetric. The final theorem states that every edge has at most 65 common neighbours and every pair of distinct non-adjacent vertices has at most 66 common non-neighbours, and it depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The graph is also checked by two independent programs, in Python and in C, which read the sets A and B and test all pairs directly. The development, the programs and the search are archived in [1].

Use of generative AI. The author used Claude (Anthropic) to search for the example, to check it by computer and in Lean, and to draft the text; the author verified every step and is responsible for the content.

References

[1] D. Bhattacharjee, *The book Ramsey number $R(B_{66}, B_{67})$: data, checks, search and Lean 4 proof*, version 1.0.0, Zenodo, 2026, doi:10.5281/zenodo.23373629. Source code at https://github.com/creelie/Breakthrough-problems-2

[2] S. Chung, W. Du and W. J. Wesley, *Autonomous mathematical discovery in an open-world multi-agent environment*, preprint, 2026. arXiv:2608.23691

[3] Epoch AI, *FrontierMath open problems: Ramsey numbers for book graphs*, 2026. https://epoch.ai/frontiermath/open-problems/ramsey-book-graphs, accessed 14 October 2026.

[4] A. W. Goodman, *On sets of acquaintances and strangers at any party*, Amer. Math. Monthly **66** (1959), no. 9, 778–783. doi:10.1080/00029890.1959.11989408

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, Cham, 2021, 625–635. doi:10.1007/978-3-030-79876-5_37

[6] L. Perron and F. Didier, *CP-SAT*, OR-Tools version 9.15, Google, 2025. https://developers.google.com/optimization/cp/cp_solver

[7] C. C. Rousseau and J. Sheehan, *On Ramsey numbers for books*, J. Graph Theory **2** (1978), 77–87. doi:10.1002/jgt.3190020110

[8] G. Szekeres, *Tournaments and Hadamard matrices*, Enseign. Math. (2) **15** (1969), 269–278.

[9] D. Turturean, *Summary of new results on the Ramsey numbers for book graphs open problem*, 2026. https://docs.google.com/document/d/1VinXOiMov2v-Y27GTJwoAECywv2a34MH4E8qa712zJY

[10] W. J. Wesley, *Lower bounds for book Ramsey numbers*, Discrete Math. **349** (2026), no. 5, 114913. doi:10.1016/j.disc.2025.114913

Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL),
Bhubaneswar, Odisha 751030, India

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: https://orcid.org/0000-0003-0466-750X